Bounce Brawl — Technical Sheet

Tech used

One self-contained HTML file (about 29 KB), plain JavaScript, HTML5 Canvas 2D with a tilted pseudo-3D arena. No libraries, images or downloads.

Web Audio API: every sound effect and the chiptune music loop are synthesized live. The music speeds up as the ring shrinks or the round goes on.

localStorage for best total, bot difficulty and mute.

Mobile first: floating virtual joystick, two large thumb buttons, safe-area insets, no zoom, fixed 60 Hz physics step.

Systems

Physics: velocity, friction, max speed, gravity and jump arcs; blob-to-blob impulse collisions where dashing multiplies knockback 1.9x and stuns.

Ground-pound shockwave on landing; bumpers; rotating beam sweepers with speed-up.

Ring-out detection, crown pickup/steal/drop, respawns, placement scoring with a double-points finale.

Bot AI: picks a target (crown holder first), gets on the centre side of it to push outward, edge-safety steering, recovery when sliding out, aligned dashes, and beam jumping using predicted beam approach.

Graphics: neon glow, gradients, squash and stretch, wobble, eyes that look where you move, stun eyes, dash trails, particles, shockwave rings, screenshake, popup text, twinkling starfield.

Audit 1: Balance (simulated play)

The full 6-round match was run headless with all-bot lobbies and with a bot-controlled player.

First version: rounds ended far too fast (Spin Sweep averaged 8.8 s, Double Sweep 6.3 s, ring rounds about 10 to 12 s). Fixes: knockback 0.62 to 0.5, dash 560 to 480, slower ring shrink, beam starts slower and off the spawn line, weaker beam swat, wider bot edge margin, more beam jumping.

After: Shrink Ring averages about 27 s, Bumper Ring 20 s, Spin Sweep 17 s, Double Sweep 14 s, crown rounds 40 s; no round stalled. Seat wins in all-bot matches spread evenly.

Difficulty: at first Chill was too hard because bots hunted the player. Now Chill bots avoid you and are slower, Normal is neutral, Spicy hunts you a little. A normal-skill simulated player wins about 75% on Chill, about 17% on Normal (top 3 in about 83%), and rarely on Spicy.

Audit 2: Bugs

Played menus, difficulty toggle, all 6 rounds, results, final screen, play again, menu and mute through a scripted DOM harness.

Fixed: the round-end state could be overwritten after the results screen opened (state is now set before the end timer). Holding dash simply re-dashes when ready (intended). Audio start is wrapped so

browsers without Web Audio still play.

Audit 3: Player experience

After falling in a knockout round, the rest of the round plays at 3x speed so you are not stuck waiting.

Shrinking rings show a pulsing red danger ring and the edge glows red when small. Popups say BYE!, CROWN! and STOLEN!; a tip tells you when you fall or hold the crown.

Clear 3-2-1-GO countdown, a rules card before each round, a standings table after each round, and buttons that grey out on cooldown.

Tested with simulated play, not on a real phone.